

	1	2	3	4	5	
A	12V Power Input  File: POS12_power_input.kicad_sch		Platform ETC  File: platform_etc.kicad_sch			A
	12V Telemetry  File: POS12_telemetry.kicad_sch		PGOOD LEDs  File: pgood_leds.kicad_sch			
	3.3V Power Supply  File: POS3P3_power_supply.kicad_sch		Status LEDs  File: status_leds.kicad_sch			
	3.3V Telemetry  File: POS3P3_telemetry.kicad_sch		RCA Signal Conditioning  File: rca_signal_conditioning.kicad_sch			
B	dsPIC33 Programming  File: dspic33_programming.kicad_sch		Speaker Signal Conditioning  File: speaker_signal_conditoning.kicad_sch			B
	dsPIC33 Power  File: dspic33_power.kicad_sch		Source Selection, PGAs  File: source_selection_pgas.kicad_sch			
	dsPIC33 GPIO  File: dspic33_gpio.kicad_sch		Anti-alias Filtering  File: anti_alias_filtering.kicad_sch			
	USB UART Bridge  File: usb_uart_bridge.kicad_sch		Audio Voltage References  File: audio_voltage_references.kicad_sch			
			LED Panel Buffers  File: led_panel_buffers.kicad_sch			
			Pushbuttons  File: pushbuttons.kicad_sch			
C			Mechanical  File: mechanical.kicad_sch			C
D						D
	1	2	3	4	5	

TRY TO TALLY LOAD CURRENTS

TP1

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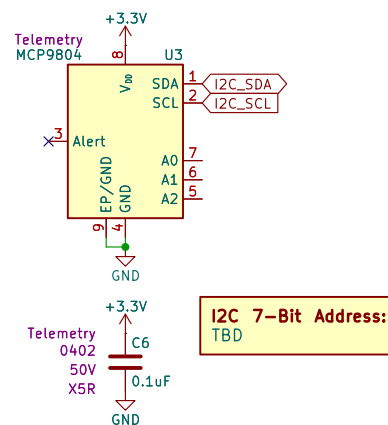
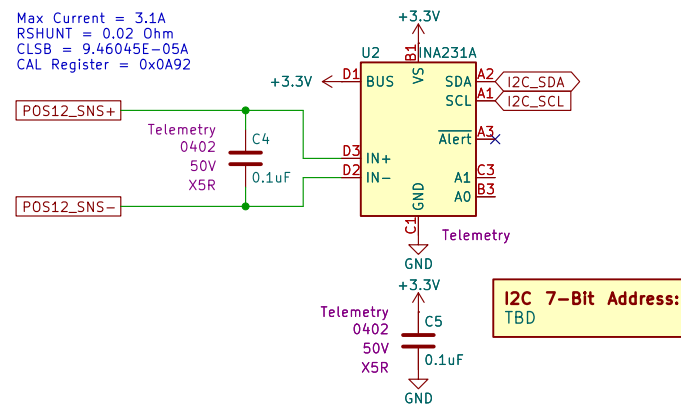
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KICad E.D.A. 10.0.3		Id: 2/20

Size: A	Date: 2023-12-25	Rev:
KiCad E.D.A. 10.0.3		Id: 2/20

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[illegible][illegible]



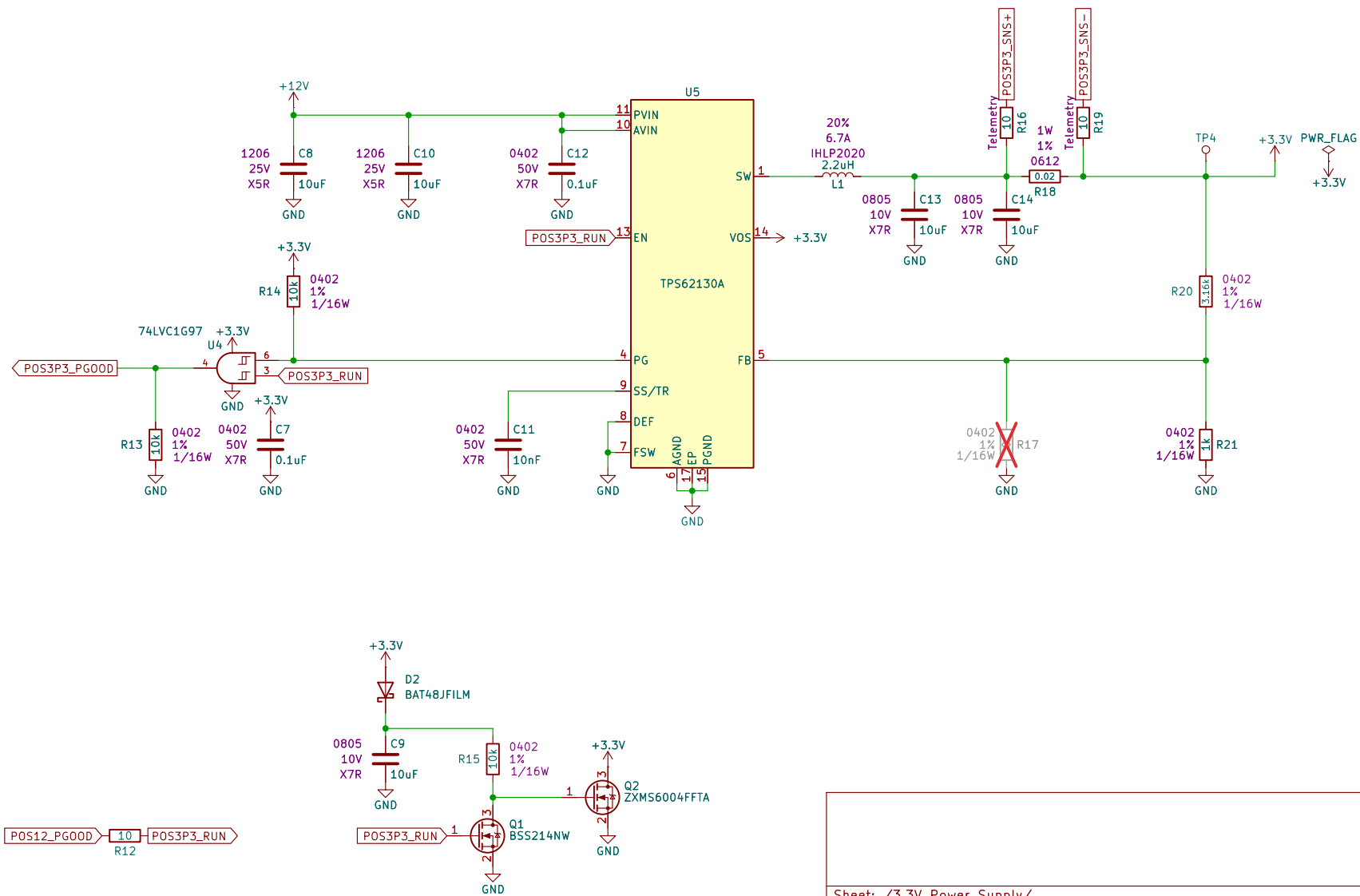
Sheet: /12V Telemetry/
 File: POS12_telemetry.kicad_sch

Title:

Size: A Date: 2023-12-25
 KiCad E.D.A. 10.0.3

Rev:
 Id: 3/20

TRY TO TALLY LOAD CURRENTS



Sheet: /3.3V Power Supply/
File: POS3P3_power_supply.kicad_sch

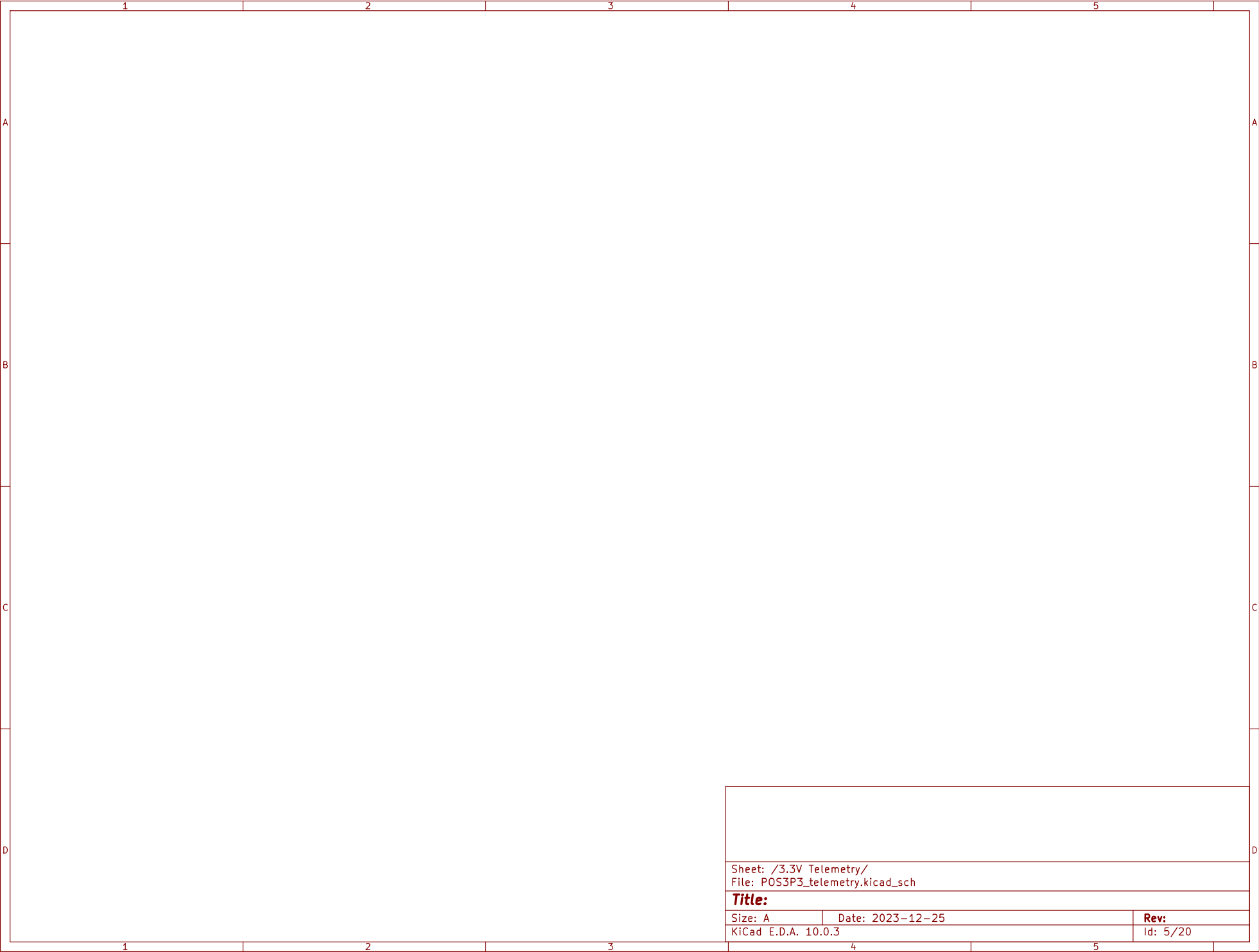
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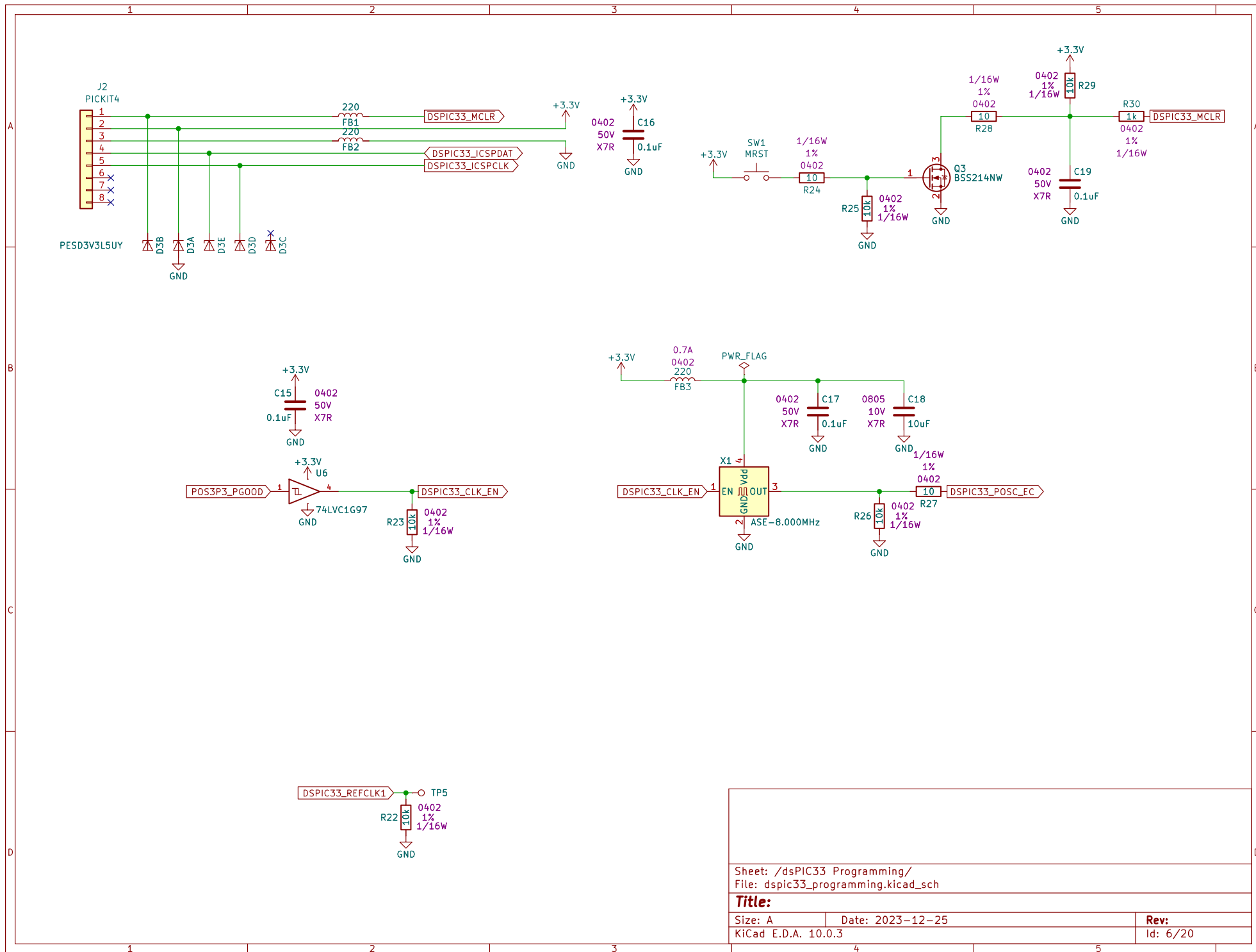
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KiCad E.D.A. 10.0.3

Rev:

Id: 4/20





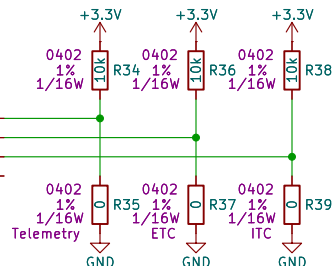
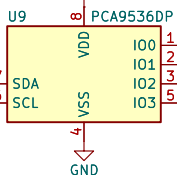
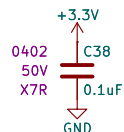
Sheet: /dsPIC33 Programming/
File: dspic33_programming.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 6/20

I2C 7-Bit Address:
0x41



U7B
dsPIC33AK256MPS105

AD1ANN2
OA1IN+/AD1AN1
CVDAN5/INT0

+1.6V_AREF_LEFT
ADC_AUDIO_LEFT
POWER_BUTTON
AUDIO_VREF_ENABLE

AD2ANN2
OA2IN+/AD2AN1
SDA1/PGD1
SCL1/PGC1

RP23 = REF01

+1.6V_AREF_RIGHT
ADC_AUDIO_RIGHT
DSPIC33_ICSPDAT
DSPIC33_ICSPCLK

DSPIC33_REFCLK1

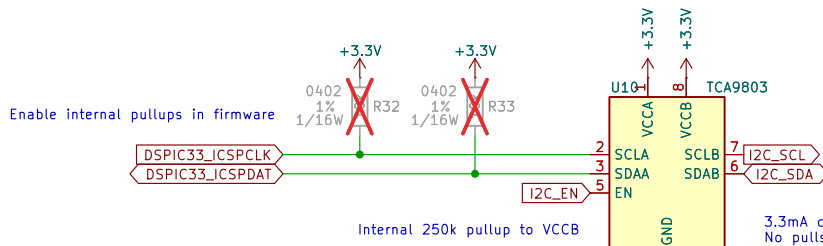
1 RA0/IOMAF2/SCL2/RP1/CVDAN0/AD1AN4/AD1ANN1/PGD2
3 RA1/RP2/CMP4D/CVDAN1/AD3AN0/AD3ANN1/DACOUT1/PGC2
8 RA2/RP3/CMP1A/CVDAN2/AD1AN0/OA1OUT
9 RA3/RP4/CMP1C/CMPCN/CVDAN3/AD1AN2/AD1ANN2/OA1IN-
10 RA4/RP5/CMP1B/CVDAN4/AD3AN4/AD2AN5/AD1AN1/OA1IN+/UREF
11 RA5/INT0/RP6/CMP3A/CVDAN5/OA3OUT
12 RA6/RP7/CMP3C/CMPCN/CVDAN6/OA3IN-
2 RA7/IOMAF1/SDA2/RP8/CVDAN7/AD3AN1
4 RA8/RP9/ISRC3/IBIAS3/CMP5A/CVDAN8/AD3AN3
5 RA9/RP10/ISRC2/IBIAS2/CMP5B/CVDAN9/AD1AN3
16 RB0/RP17/CMP2A/CVDTX0/CVDAN16/AD2AN0/OA2OUT
17 RB1/RP18/CMP2C/CVDTX1/CVDAN17/AD2AN2/AD2ANN2/OA2IN-/TMS
18 RB2/RP19/CMP2B/CVDTX2/CVDAN18/AD2AN1/OA2IN+
21 RB3/SDA1/RP20/CMP4A/CVDTX3/CVDAN19/AD2AN3/AD2ANN1/PGD1
22 RB4/SCL1/RP21/CMP4B/CVDTX4/CVDAN20/AD2AN4/PGC1
13 RB5/RP22/CMP3B/CVDTX5/CVDAN21/OA3IN+
23 RB6/I3CASCL1/RP23/ISRC0/IBIAS0/ALLCMP/CVDTX6/CVDAN22/AD3AN2/AD3ANN2
24 RB7/IOMAF0/I3CASDA1/RP24/ISRC1/IBIAS1/CMP4C/CVDTX7/CVDAN23
19 RB8/RP25/CMP2D/CVDTX8/CVDAN24
20 RB9/RP26/CVDTX9/CVDAN25

OSC0/CLK0/RP33/IOMAF5/RC0
OSCI/RP34/IOMAF6/RC1
PGC3/RP35/PWM4H/IOMAD7/RC2
RP36/PWM3H/IOMAD5/RC3
RP37/I3CSDA1/PWM3L/IOMAD4/RC4
PGD3/RP38/I3CSCL1/PWM4L/IOMAD6/RC5
RP39/ASCL1/RC6
RP40/ASDA1/IOMAF7/RC7
RP49/PWM2H/IOMAD3/RD0
TCK/RP50/PWM2L/IOMAD2/RD1
TDO/RP51/PWM1H/IOMAD1/RD2
TDI/RP52/PWM1L/IOMAD0/RD3
RP55/RD6

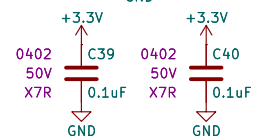
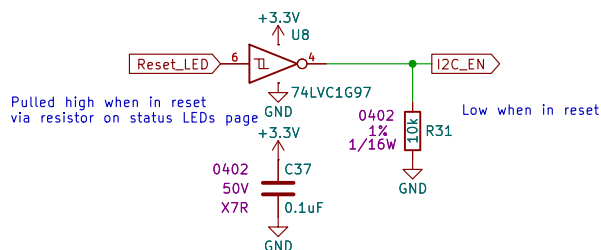
29 USB_UART_RX
30 DSPIC33_POSC_EC
31 USB_UART_TX
33 LED_DRIVER_BLANK_PWM
34 HEARTBEAT_LED
32 LED_DRIVER_LAT
25
26
37 LEFT_LED_DRIVER_SPL_SCK
38 LEFT_LED_DRIVER_SPL_SDO
39 RIGHT_LED_DRIVER_SPL_SCK
40 RIGHT_LED_DRIVER_SPL_SDO
47

RP33 = U1RX
OSCI
RP35 = U1TX
RP36 = OCM2
RP37 = OCM1

RP49 = SCK1
RP50 = SDO1
RP51 = SCK2
RP52 = SDO2



Isolate I2C signals when in reset/being programmed
so that ICSP can program the DSC but cannot
spook I2C devices with nonsensical data



AUDIO_SOURCE_SELECT

POT_RESET

RESET_LED

PGOOD_LED_SHDN

ERROR_LED

CPU_TRAP_LED

Sheet: /dsPIC33 GPIO/
File: dspic33_gpio.kicad_sch

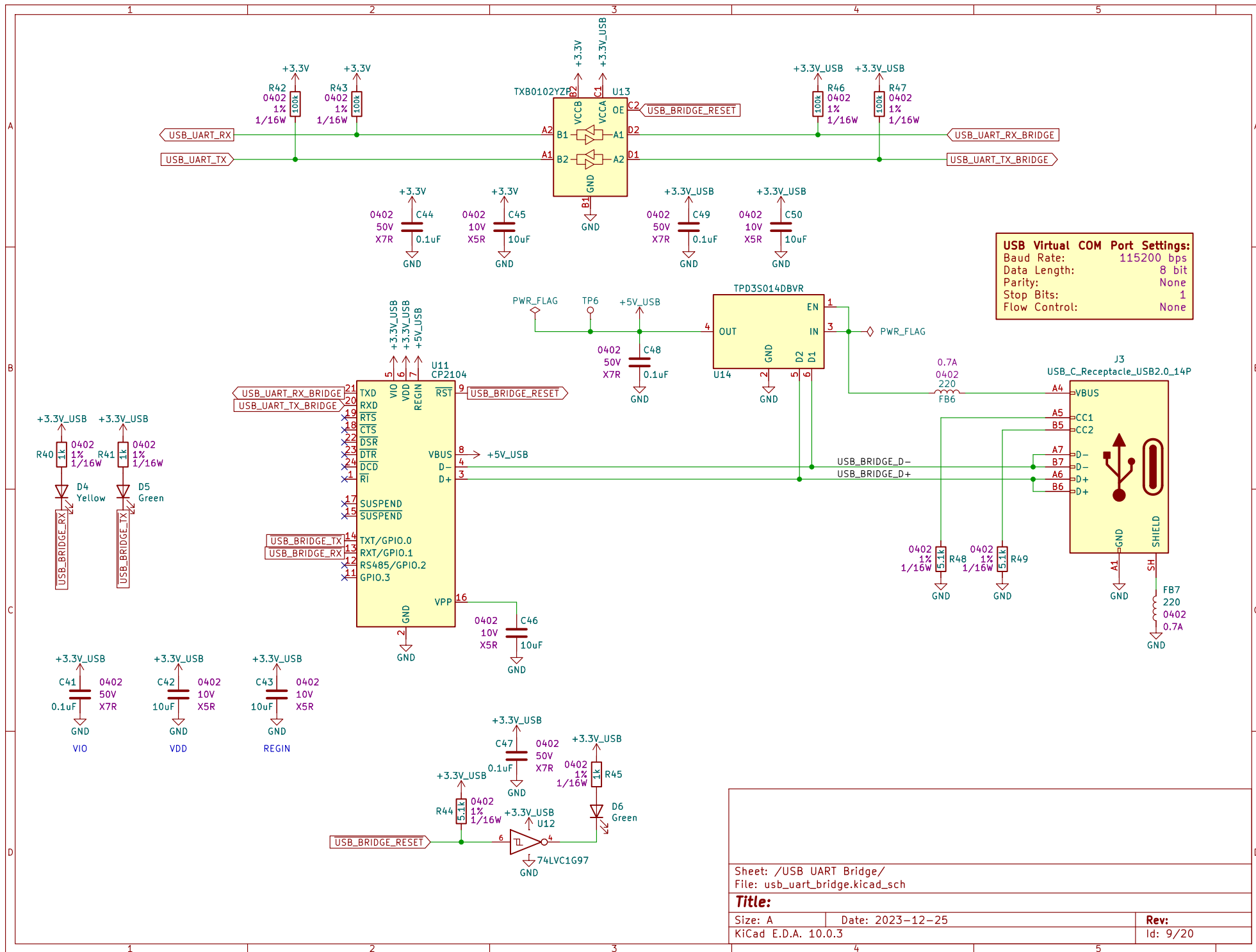
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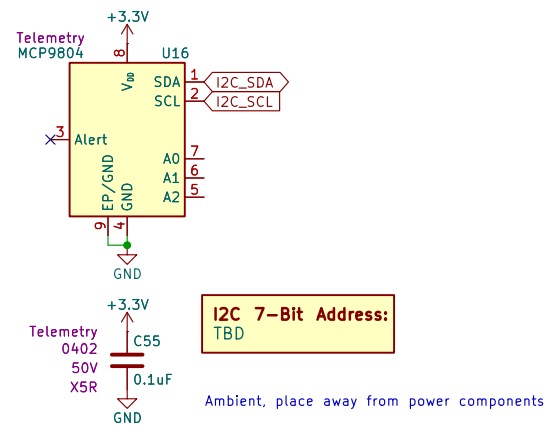
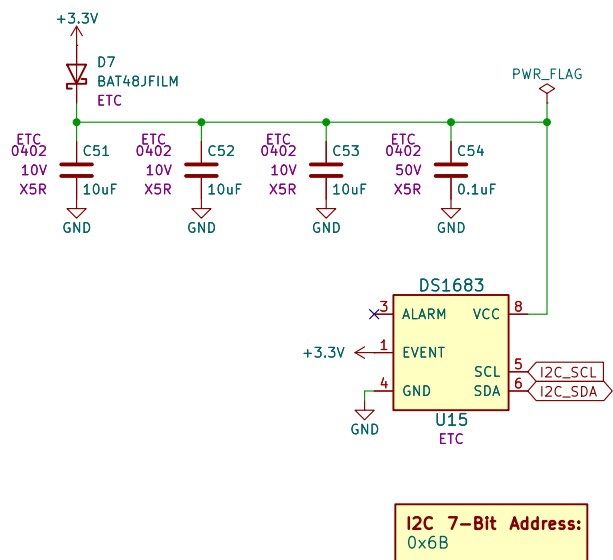
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KiCad E.D.A. 10.0.3

Rev:

Id: 8/20



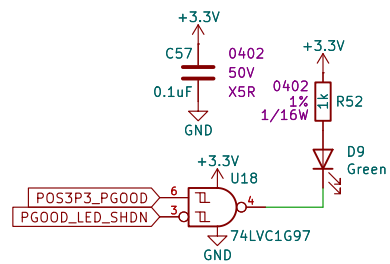
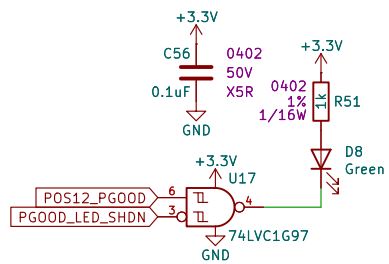
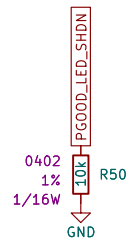


Sheet: /Platform ETC/
File: platform_etc.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 10/20

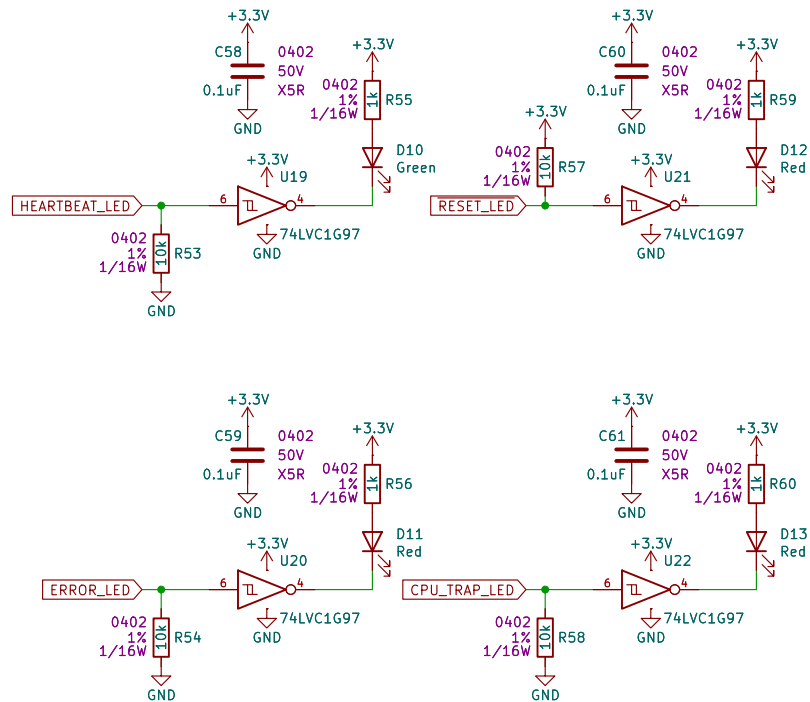


Sheet: /PGOOD LEDs/
File: pgood_leds.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 11/20



Sheet: /Status LEDs/
File: status_leds.kicad_sch

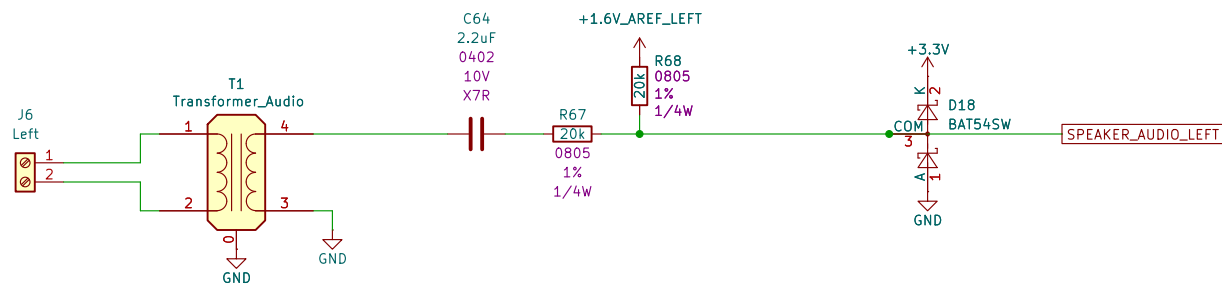
Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 12/20

Rev:
Id: 13/20

Alternate is a 20:1 resistive attenuation (for 18V peaks) and a diff amp



Sheet: /Speaker Signal Conditioning/
File: speaker_signal_conditioning.kicad_sch

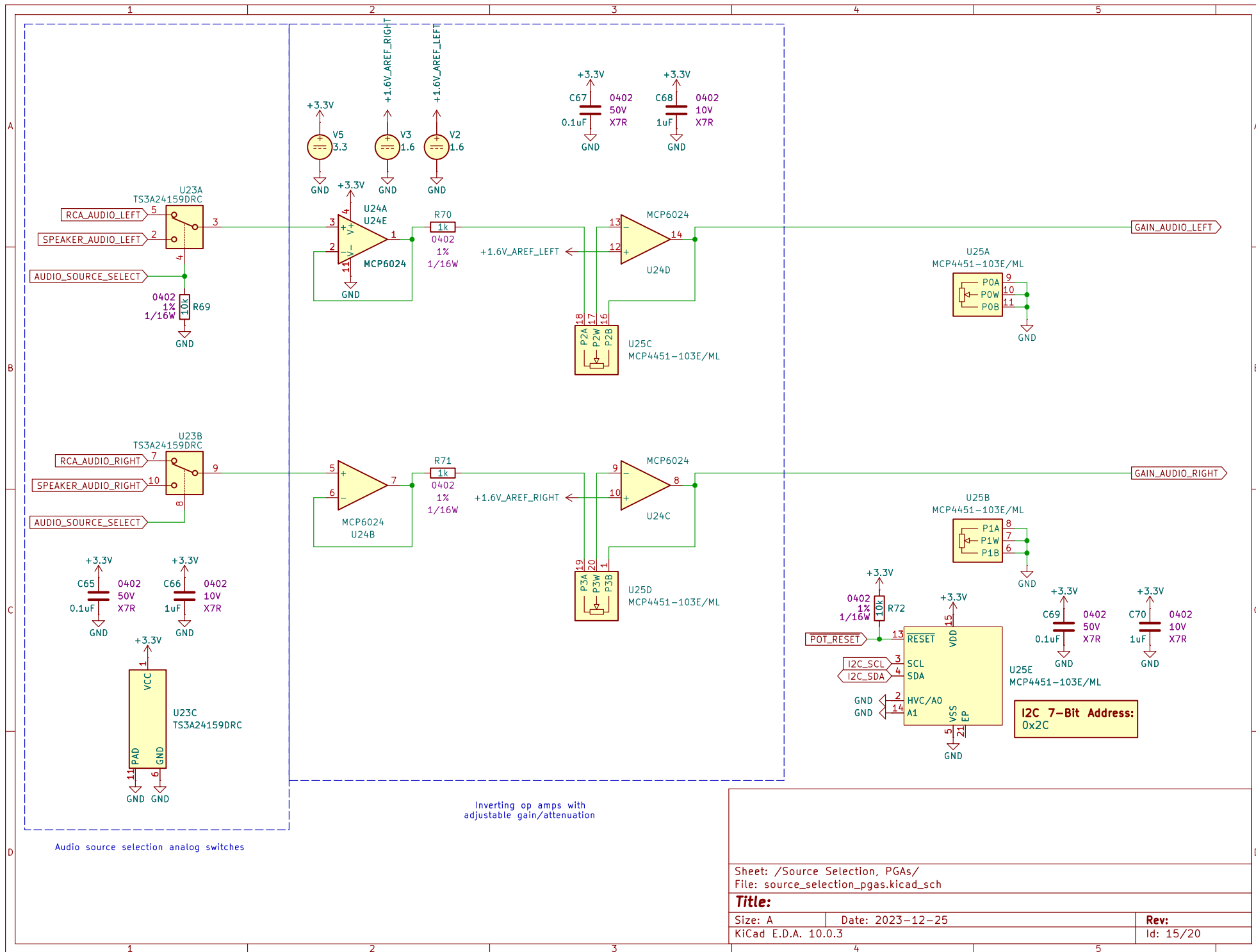
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Size: A Date: 2023-12-25

KiCad E.D.A. 10.0.3

Rev:

Id: 14/20

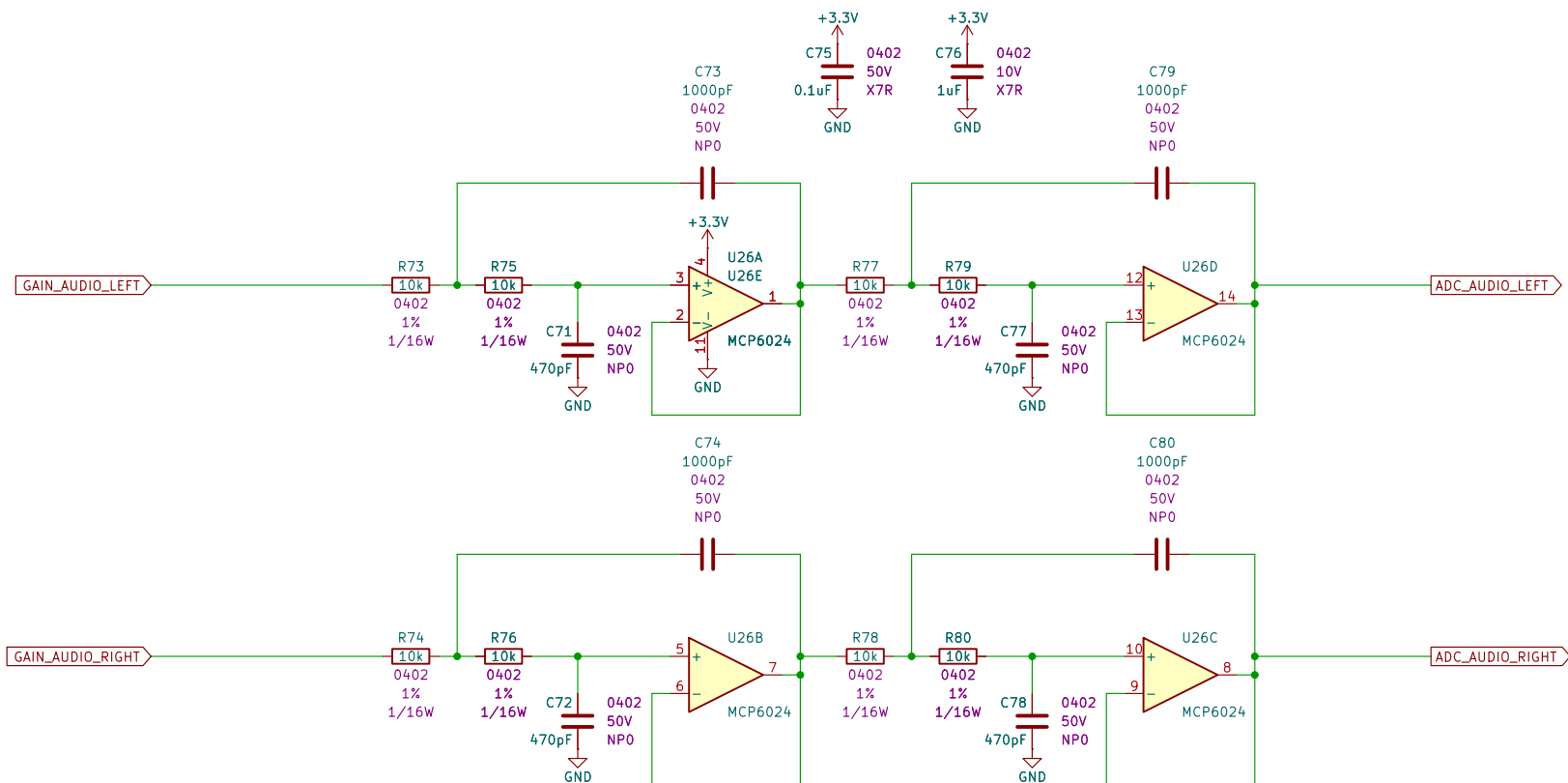


Sheet: /Source Selection, PGAs/
File: source_selection_pgas.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 15/20



4-pole Sallen-Key active anti-aliasing lowpass filters
 Unity gain
 $f_c = 23.2\text{kHz}$, $Q = 0.707$ each
 If sampling at 44.1kHz , nyquist is 22.05kHz
 Must filter hard just below this

Sheet: /Anti-alias Filtering/
 File: anti_alias_filtering.kicad_sch

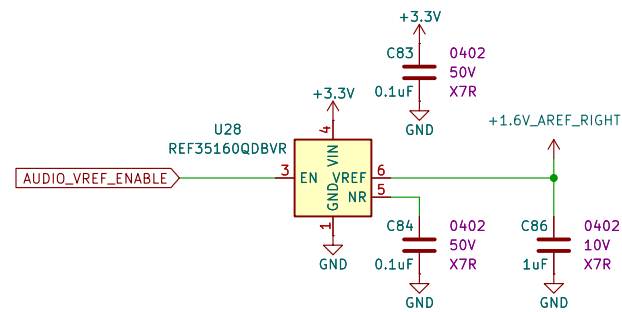
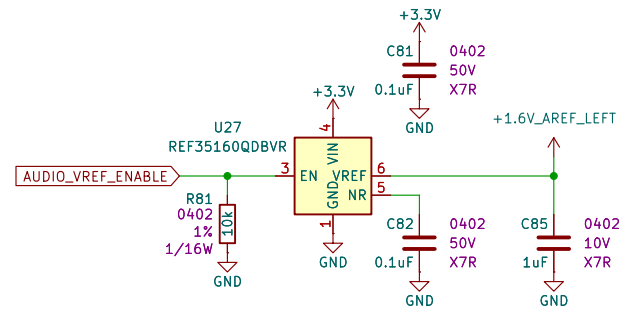
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Size: A Date: 2023-12-25

KiCad E.D.A. 10.0.3

Rev:

Id: 16/20

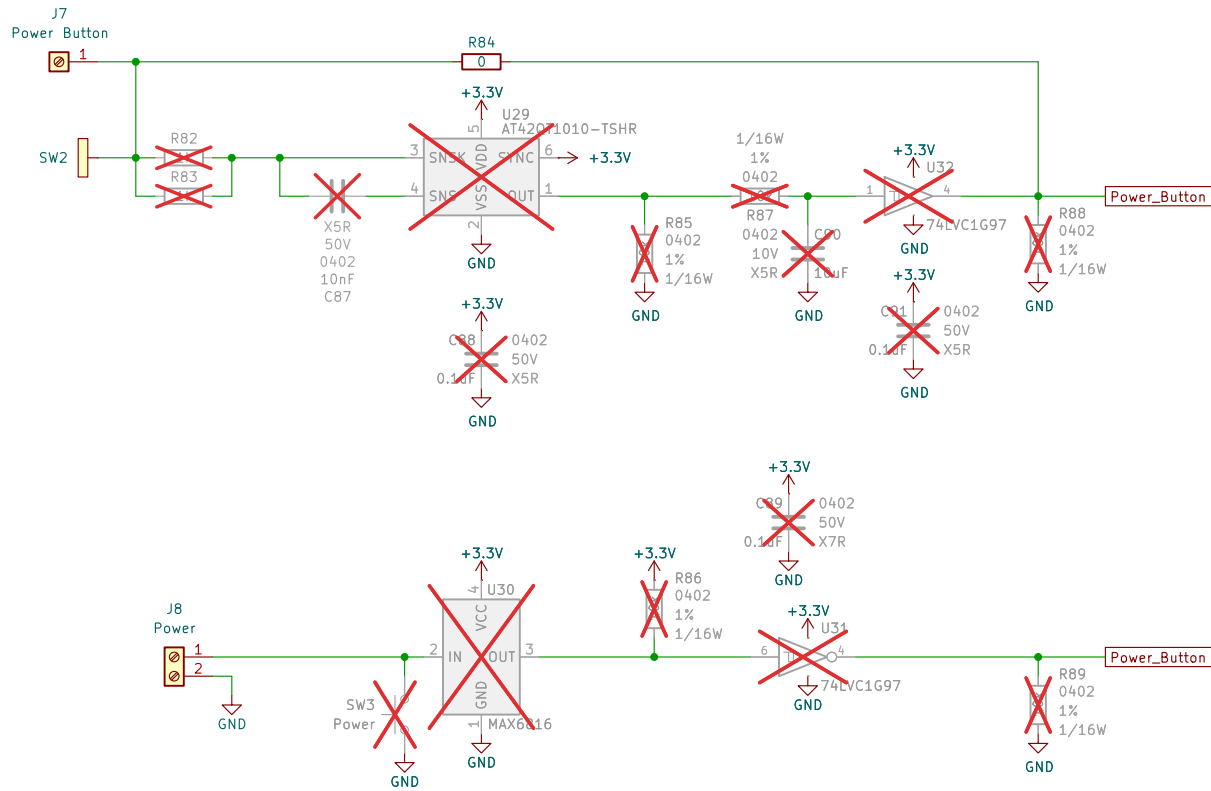


Sheet: /Audio Voltage References/
File: audio_voltage_references.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 17/20



Sheet: /Pushbuttons/		
File: pushbuttons.kicad_sch		
Title:		
Size: A	Date:	Rev:
KiCad E.D.A. 10.0.3		Id: 19/20

